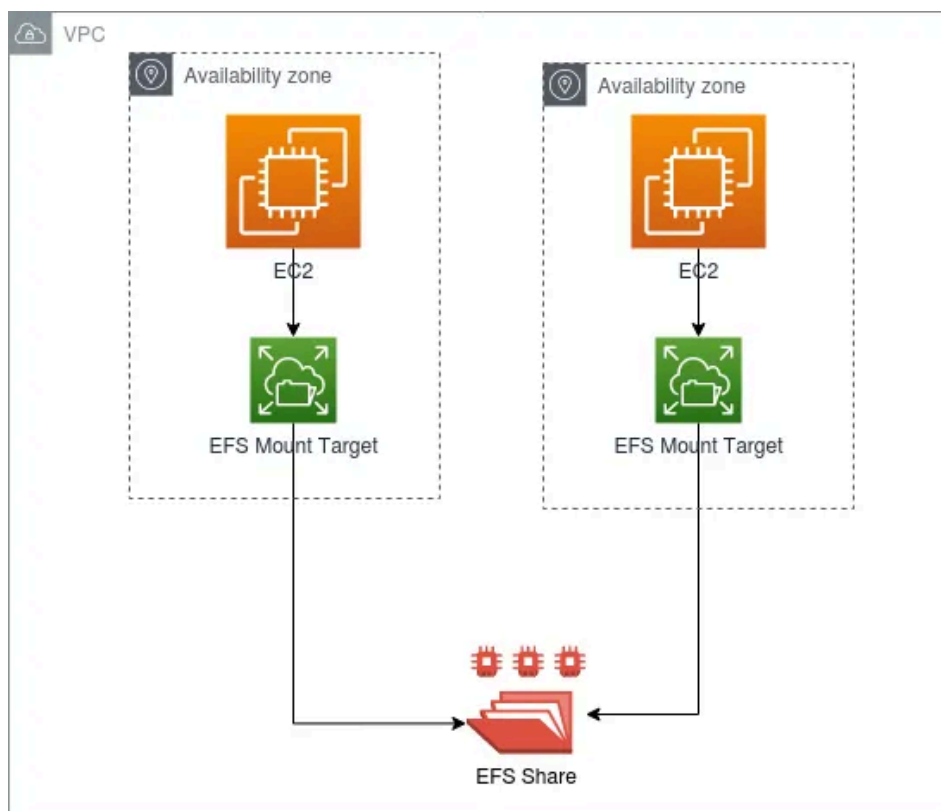


AWS-EFS (Files Sharing System)

What is Aws Elastic File System(EFS)

Amazon Elastic File System (EFS) is a serverless, fully managed, elastic file system that lets you share file data without provisioning or managing storage. It can be used with AWS Cloud services and on-premises resources, and is built to scale on demand to petabytes without disrupting applications.

Unlike EBS (which is like a local hard drive for one server), EFS is a Network File System (NFS). This means hundreds or thousands of EC2 instances, Lambda functions, and Containers can access the same files at the same time.



Amazon EFS (Elastic File System) – Simple Explanation

How EFS Works

- Amazon EFS is a fully managed file storage service
- It is created in a region and automatically replicated across multiple Availability Zones for high availability
- You can mount EFS on multiple EC2 instances at the same time

- Uses NFS protocol, so it works like a normal folder
(example: `/mnt/efs`)
 - Accessible through mount targets created inside your VPC
-

Use Cases of Amazon EFS

- ✓ Secure file sharing – Share files across multiple instances with consistency
 - ✓ Web hosting – Multiple web servers can access the same files
 - ✓ Modern applications – Works with EC2, ECS, EKS, and serverless apps
 - ✓ ML & AI workloads – Ideal for large datasets accessed by many systems
-

Performance & Throughput Modes

Performance Modes

- General Purpose – Low latency, best for web apps & file sharing
- Max I/O – High throughput for big data and parallel workloads

Throughput Modes

- Elastic (Recommended) – Automatically scales, pay for what you use
- Bursting – Good for small workloads with traffic spikes
- Provisioned – Fixed throughput for high-speed needs with small data

Follow the Steps Properly...

```
EC2 Dashboard>Network & Security>
Security groups>create ...
Security group name :-webserve.SG5
Description :-nfs
VPC :-default vpc
Inbound Rules Edit
Type :-All traffic
Customer :-0.0.0.0
...create
```

EC2 > Security Groups > Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
All traffic	All	All	Anywhere...	nfs	Delete
0.0.0.0/0					

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info	
All traffic	All	All	Custom		Delete
0.0.0.0/0					

Second Security groups create

Security group name :-security-group

Description :-nfs

VPC :-default vpc

Inbound Rules Edit

Type :-All traffic

Customer :-0.0.0.0

...create

EC2 > Security Groups > Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

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Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
All traffic	All	All	Anywhere...		Delete
0.0.0.0/0					

[Add rule](#)

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Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info	
All traffic	All	All	Custom		Delete
0.0.0.0/0					

Search EC2>Launch instance

Name :-linux-1

Image :-aws

Type :-t3.micro

Key-pair :-practise.pem

Name
[Add additional tags](#)

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

aws

Mac

ubuntu

Microsoft

Red Hat

SUSE

debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-02ed5a24b218ff2e (64-bit (x86), uefi-preferred) / ami-01605a62a13170ae53 (64-bit (ARM), uefi)
Virtualization: hvm (ENI-enabled true) Boot device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260105.0 x86_64 HVM kernel-6.1

Architecture

Boot mode

AMI ID

Publish Date

Username

64-bit (x86)

uefi-preferred

ami-0ced5a24b218ff2e

2026-01-02

ec2-user

Verified provider

Instance type

t3.micro

Family t3 2 vCPU 1 GB Memory Current generation true On Demand Linux base pricing: 0.0112 USD per Hour On Demand SUSE base pricing: 0.0113 USD per Hour On Demand Windows base pricing: 0.0204 USD per Hour On Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

practise-key

Create new key pair

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...read more
ami-02ed5a24b218ff2e

Virtual server type (instance type)

t3.micro

Firewall (security group)

webserveS.G01

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Edit network

VPC - *required* :- default vpc

Subnet :-1a

Availability Zone :-ap-south-asia-1a

Firewall :-select existing security group

Common security group :- select webserver-S.G01

...create

Network settings

VPC - required

Info

vpc-060d64bd6f8be855

(default)

172.31.0.0/16

Subnet

Info

subnet-050e2ce75ca6f9e98

VPC: vpc-060d64bd6f8be855 Owner: 601920987231 Availability Zone: ap-south-1c (aps1-az2) Zone type: Availability Zone IP addresses available: 4091 CIDR: 172.31.16.0/20

Create new subnet

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

Info

Select security groups

webserveS.G01 sg-00a4122e1546b3c3f

VPC: vpc-060d64bd6f8be855

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

Name :-linux-2

Image :-aws

Type :-t3.micro

Key-pair :-practise.pem

Name: linux1

Application and OS Images (Amazon Machine Image)

Amazon Linux 2023 (kernel-6.1)

Instance type: t3.micro

Key pair (login): practise-key

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...

Virtual server type (Instance type): t3.micro

Firewall (security group): webserver-SG01

Storage (volumes): 1 volume(s) - 8 GiB

Launch Instance

Edit network

VPC - *required* :- default vpc

Subnet :-1a

Availability Zone :-ap-south-asia-1a

Firewall :-select existing security group

Common security security group :- select webserver-S.G01

...create

Network settings

VPC - *required*: vpc-060d64bd6f8be855 (default)

Subnet: subnet-087db0ba2d084552b

Auto-assign public IP: Enable

Firewall (security groups): Select existing security group

Common security groups: webserver-SG01

Search efs>customize>create

Create file system

Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

Name - optional
Name your file system.

Name can include letters, numbers, and +, -, _, / symbols, up to 256 characters.

Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.

default

Recommended settings
Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#).

Setting	Value	Editable after creation
Throughput mode Learn more	Elastic	Yes
Transition into Infrequent Access (IA)	30 day(s) since last access	Yes
Transition into Archive	90 day(s) since last access	Yes
Transition into Standard	None	Yes
Automatic backups	Enabled	Yes
Encryption	Enabled	No

[Cancel](#) [Customize](#) [Create file system](#)

Name - optional :-magicbuss-efs-test

File system type :-Regional zone(provide multiply availability zone at once)

Automatic backups :-uncheck

Encryption :-uncheck

Performance settings

Throughput mode :-Enhanced

Check :- Elastic (Recommended)

Additional setting:- General Purpose (Recommended)

....next

Automatic backups

Automatically backup your file system data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☐ Enable automatic backups

 We recommend that you create a backup policy for your file system

Lifecycle management

Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)

Transition files to IA based on the time since they were last accessed in Standard storage.

30 day(s) since last access

Transition into Archive

Transition files to Archive based on the time since they were last accessed in Standard storage.

90 day(s) since last access

Transition into Standard

Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.

None

Encryption

Choose to enable encryption of your file system's data at rest. Uses the AWS KMS service key (aws/elasticfilesystem) by default. [Learn more](#)

☐ Enable encryption of data at rest

Performance settings

Throughput mode

Choose a method for your file system's throughput limits. [Learn more](#)

☒ Enhanced

Provides more flexibility and higher throughput levels for workloads with a range of performance requirements.

☐ Bursting

Provides throughput that scales with the amount of storage for workloads with basic performance requirements.

☒ Elastic (Recommended)

Use this mode for workloads with unpredictable I/O. With Elastic Throughput, performance automatically scales with your workload activity and you only pay for the throughput you use (data transferred for your file systems per month). [Learn more](#)

☐ Provisioned

Use this mode if you can estimate your workload's throughput requirements. With Provisioned mode, you configure your file system's throughput and pay for throughput provisioned.

▼ Additional settings

Performance mode

Set your file system's performance mode based on IOPS required. File systems using Elastic throughput mode only support General Purpose performance mode. [Learn more](#)

☒ General Purpose (Recommended)

Ideal for a variety of diverse workloads, including high performance and latency-sensitive applications

☐ Max I/O

Designed for highly parallelized workloads that can tolerate higher latencies

Network access

Network vpc :-default

Mount targets

Availability zone :- ap-south-asia-1a

Ip address type :-ipv4 only security group :-webserver

Network access

Network
Virtual Private Cloud (VPC) [Learn more](#)
Choose the VPC where you want EC2 instances to connect to your file system.
vpc-060d64bd6f6be855
default

Mount targets
A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone	Subnet ID	IP address type	IPv4 address	IPv6 address	Security groups	
ap-south-1a	subnet-053...	IPv4 only	Optional	-	sg-00a4122e1546b3c3f webserve5.G01	Remove
ap-south-1b	subnet-087...	IPv4 only	Optional	-	sg-00a4122e1546b3c3f webserve5.G01	Remove

Add mount target

Cancel Previous Next

Availability zone :- ap-south-asia-1a
Ip address type :-ipv4 only security group :-webserver

Attach

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS

☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-00b3b5b8d6d538931:/ efs
```

Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-00b3b5b8d6d538931.efs.us-east-1.amazonaws.com:/ efs
```

next> ...create
Open ec2 instance

Connect the EC1 & EC2 Instance

EC2 > Instances > i-0dabda8745ef9d7e3 > Connect to instance

Connect info
Connect to an instance using the browser-based client.

Instance ID i-0dabda8745ef9d7e3 (linux1)	VPC ID vpc-060d64bd6f6be855	Security groups sg-00a4122e1546b3c3f (webserve5.G01)	IAM role -
--	---------------------------------------	--	----------------------

EC2 Instance Connect **SSM Session Manager** **SSH client** **EC2 serial console**

Instance ID
i-0dabda8745ef9d7e3 (linux1)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is practise-key.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 "practise-key.pem"
4. Connect to your instance using its Public DNS:
ec2-65-1-180-251.ap-south-1.compute.amazonaws.com

Command copied

```
ssh -i "practise-key.pem" ec2-user@ec2-65-1-180-251.ap-south-1.compute.amazonaws.com
```

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Open ctrl+r cmd > cd / downloads >ssh paste...

```
C:\Users\Nikhil>cd Downloads

C:\Users\Nikhil\Downloads>ssh -i "practise-key.pem" ec2-user@ec2-65-1-180-251.ap-south-1.compute.amazonaws.com

#
#####
~\_#####\      Amazon Linux 2023
nn\_#####\
nn\_###|
nn\_#/
nn\_V~'!_--> https://aws.amazon.com/linux/amazon-linux-2023
nnn
nn.-./-/
_/m/'-/

ec2-user@ip-172-31-28-190 ~]$ ls
```

Open second instance two

```
C:\Users\Nikhil>cd Downloads

C:\Users\Nikhil\Downloads>ssh -i "practise-key.pem" ec2-user@ec2-65-1-180-251.ap-south-1.compute.amazonaws.com

#_
~\_ ##### Amazon Linux 2023
~~ \#####\
~~  \###|
~~   \|
~~    #/_ --- https://aws.amazon.com/linux/amazon-linux-2023
~~     V~' '->
      ~~~
        ~~-
          _.-_-/_
            _/_-'
              _/m/'
```

Command on linux

- Sudo su
- Yum install https -y

```
ec2-user@ip-172-31-28-190 ~]$ sudo su
root@ip-172-31-28-190 ec2-user[# yum install httpd -y
amazon Linux 2023 Kernel Livepatch repository
dependencies resolved.
```

Package	Architecture	Version
Installing:		
httpd	x86_64	2.4.66-1.amzn2023.0.1
Installing dependencies:		
apr	x86_64	1.7.5-1.amzn2023.0.4
apr-util	x86_64	1.6.3-1.amzn2023.0.2
apr-util-lmdb	x86_64	1.6.3-1.amzn2023.0.2

- Yum install amazon-efs-utils

```

root@ip-172-31-28-190/home x + v
Available Versions:
Version 2023.10.20260120:
Run the following command to upgrade to 2023.10.20260120:

dnf upgrade --releasever=2023.10.20260120

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10.20260120.html
Version 2023.10.20260120:
Run the following command to upgrade to 2023.10.20260120:

dnf upgrade --releasever=2023.10.20260120

Release notes:
https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10.20260120.html
=====
Installed:
amazon-efs-utils-2.4.1-1.amzn2023.x86_64          stunnel-5.58-1.amzn2023.0.2.x86_64

Complete!
root@ip-172-31-28-190 ec2-user]#

```

- Systemctl start httpd
- Mkdir/ -t efs -o tls fs-07cbe685d2340b60: /efs
- cd/ efs
- Touch a a1 a2 a3
- Touch b1 b2 b3

View the list on second ec2 the list will be same as update with the work as process

```

root@ip-172-31-12-17/efs x + v
[ root@ip-172-31-12-17 efs]# ls
a a1 a2 a3
[ root@ip-172-31-12-17 efs]# ls
a a1 a2 a3 b1 b2 b3
[ root@ip-172-31-12-17 efs]# |

root@ip-172-31-12-17/efs x + v
[ root@ip-172-31-12-17 efs]# ls
a a1 a2 a3
[ root@ip-172-31-12-17 efs]# touch b1 b2 b3
[ root@ip-172-31-12-17 efs]#

```

Sudo amount -t efs -o tls fs-07cbe685d2340b60: / /var/www/html

Vi index.html press i>insert mode {paste a html code} :wq(write and quite)

list the files on second Ec2 Instance

If the file is view the mount is working properly...

```

root@ip-172-31-12-17/efs x + v
[ root@ip-172-31-12-17 efs]# ls
a a1 a2 a3
[ root@ip-172-31-12-17 efs]# ls
a a1 a2 a3 b1 b2 b3
[ root@ip-172-31-12-17 efs]# sudo mount -t efs -o tls fs-07cbe685d2340b60:/
/var/www/html
[ root@ip-172-31-12-17 efs]# sudo mount -t efs -o tls fs-07cbe685d2340b60:/
/var/www/html
/var/www/html is already mounted, please run 'mount' command to verify
[ root@ip-172-31-12-17 efs]# vi index.html
[ root@ip-172-31-12-17 efs]#

root@ip-172-31-12-17/efs x + v
[ root@ip-172-31-12-17 efs]# ls
a a1 a2 a3 b1 b2 b3 index.html
[ root@ip-172-31-12-17 efs]# |

```

